



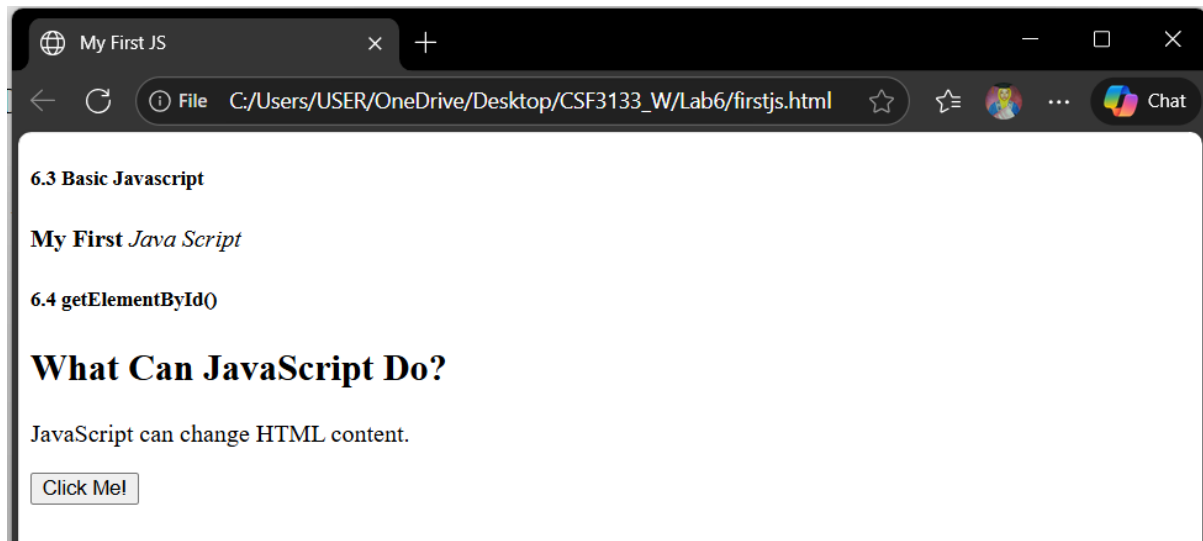
CSF3133

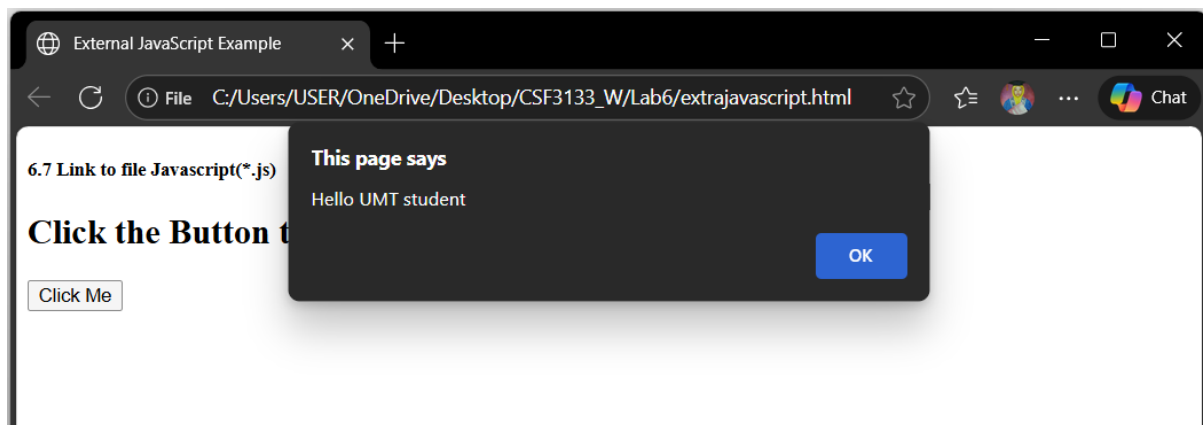
WEB-BASED INTERFACE DESIGN

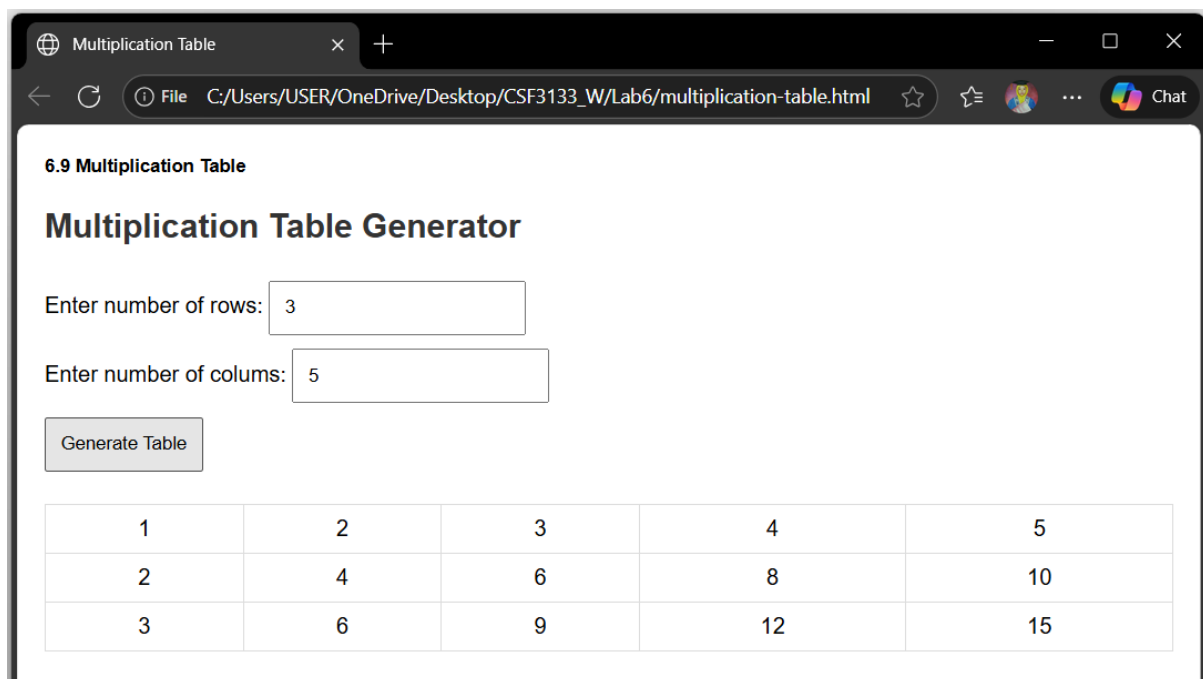
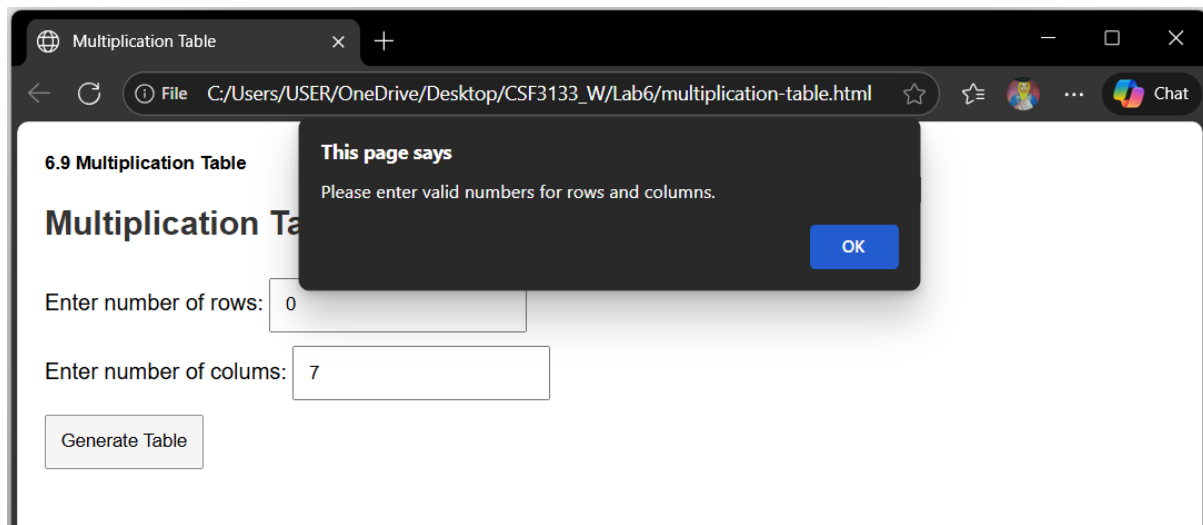
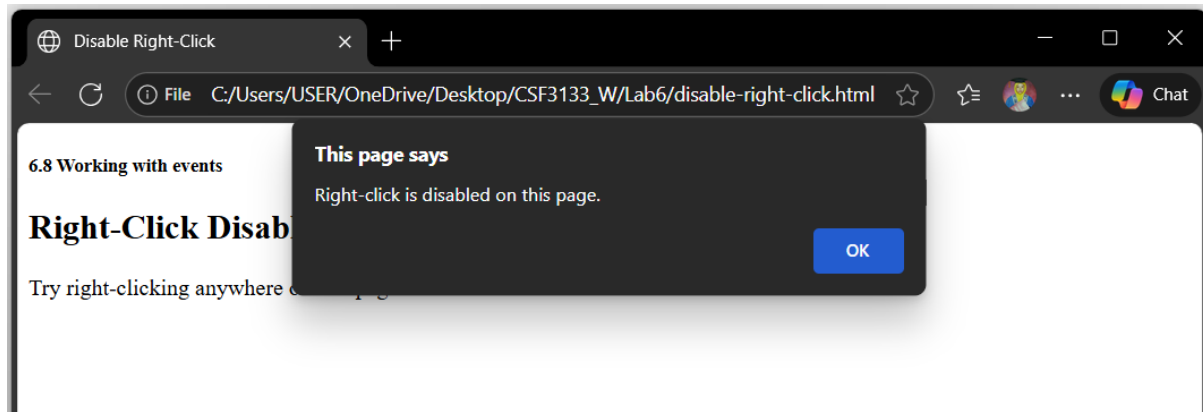
Lab Module 6

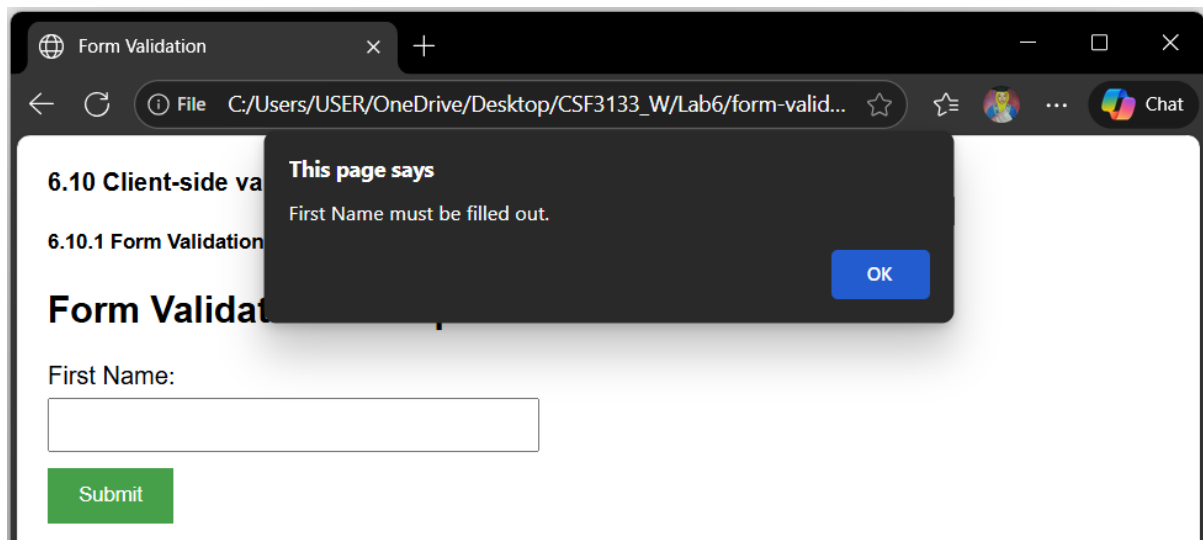
Name	: NUR MAS QASRINA BINTI MAS JAYA
Matric Number	: S74706
Date	: 3 December 2025
Group	: K4
Lecturer	: WAN NURAL JAWAHIR HJ WAN YUSSOF
Lab Moderator	: WAN FATIN FATIAH BINTI YAHYA

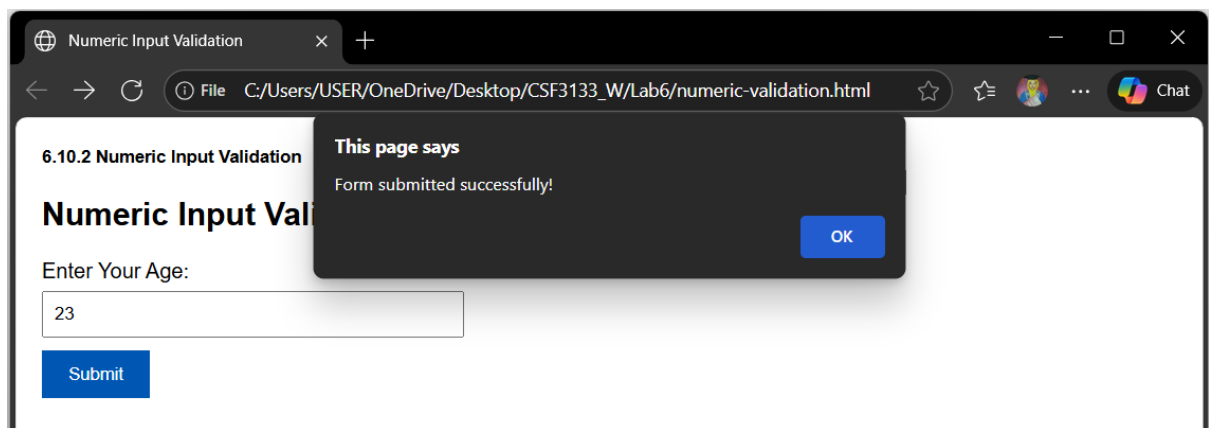
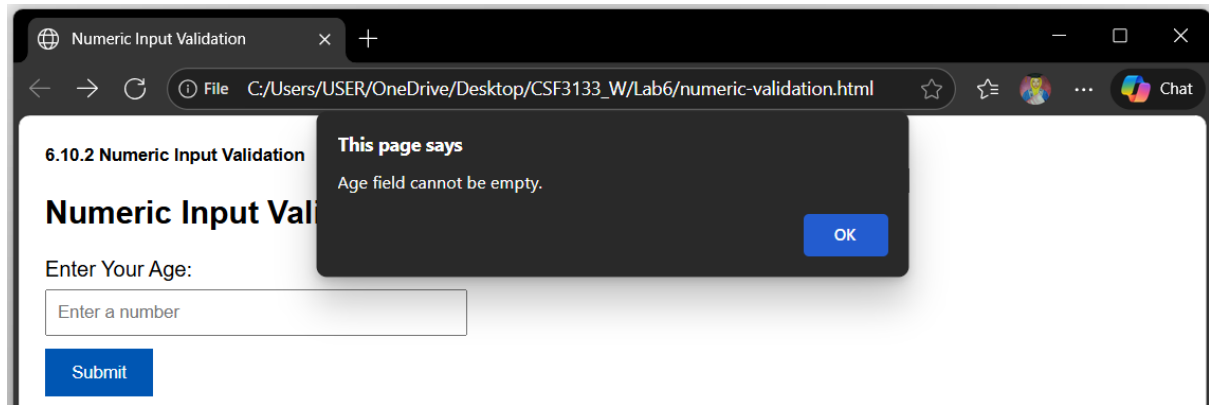
Lab Practice

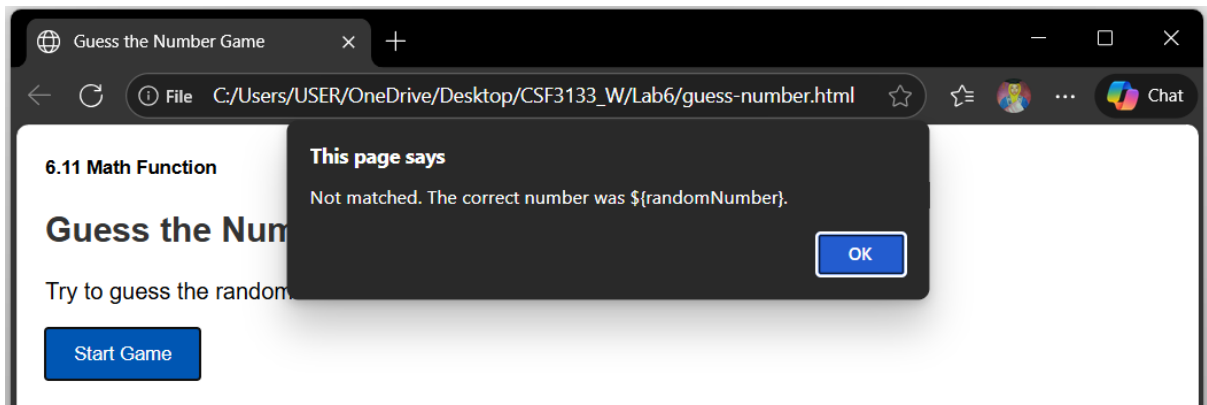
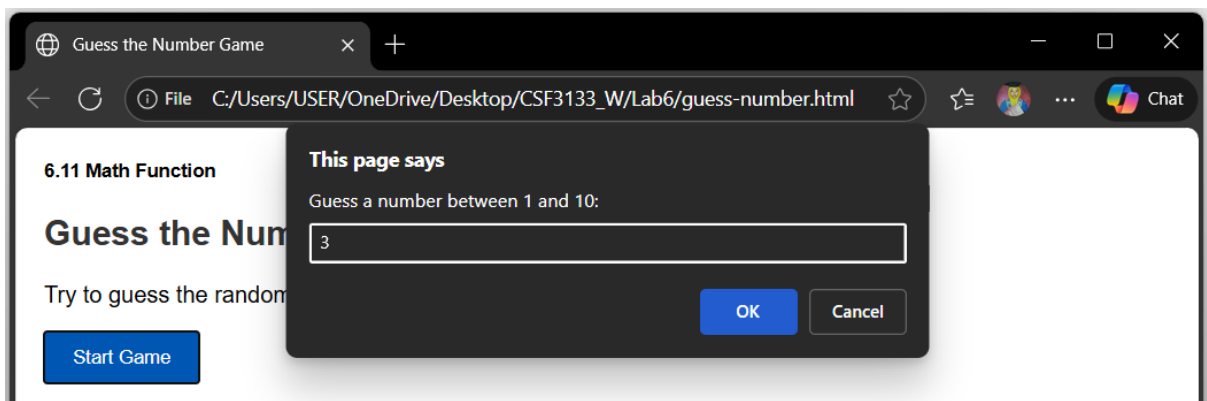
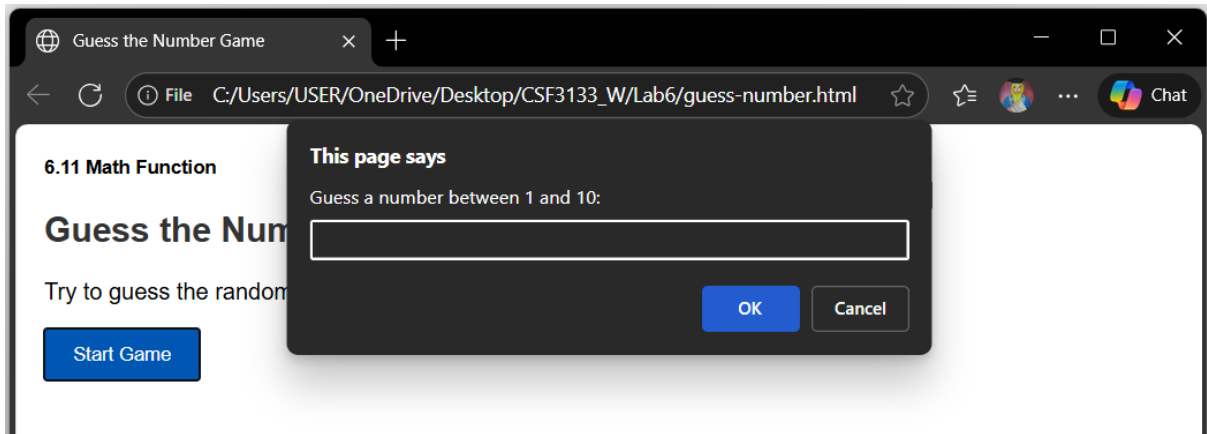


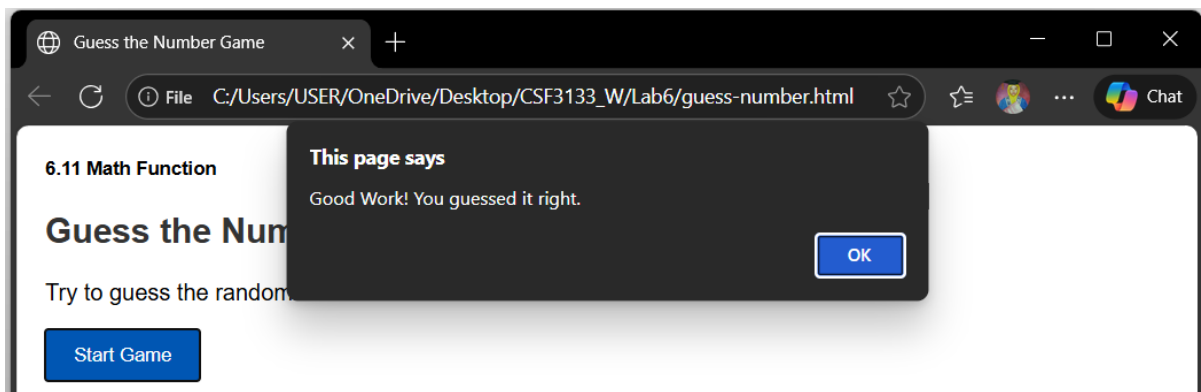
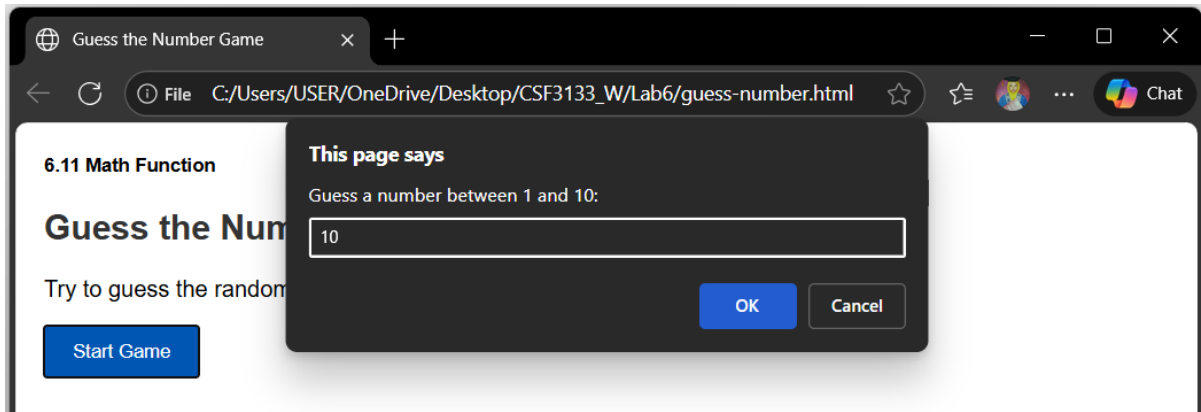












Lab Exercise 6

Index.html

```
<? index.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Dynamic Quiz Application</title>
7      <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10     <header>
11         <h1>Welcome to the General Knowledge Quiz!</h1>
12     </header>
13
14     <main>
15         <section id="quiz-section">
16             <div id="timer-container">
17                 Time Left: <span id="time">15</span>s
18             </div>
19             <div id="question-container">
20                 <h2 id="question">Question text goes here</h2>
21                 <div id="answer-buttons" class="btn-grid">
22                     </div>
23             </div>
24         </section>
25
26         <section id="feedback-section">
27             <p id="feedback-text"></p>
28         </section>
29
30         <div class="controls">
31             <button id="next-btn" class="btn hide">Next Question</button>
32         </div>
33     </main>
34
35     <footer>
36         <p id="score-container" class="hide">Your Final Score: <span id="score">0</span> / <span id="total-questions">0</span></p>
37     </footer>
38
39     <script src="script.js"></script>
40 </body>
41 </html>
```

style.css

```
# style.css > {} @media (max-width: 500px)
1  /*basic reset and body styles*/
2  * {
3      box-sizing: border-box;
4      margin: 0;
5      padding: 0;
6      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
7  }
8
9  body {
10     background-color: #f0f2f5;
11     display: flex;
12     flex-direction: column;
13     min-height: 100vh;
14     align-items: center;
15     padding-top: 2rem;
16 }
17
18 /*header*/
19 header {
20     margin-bottom: 1.5rem;
21     text-align: center;
22     color: #333;
23 }
24
25 /*main quiz*/
26 main {
27     background-color: #fff;
28     padding: 2rem;
29     border-radius: 12px;
30     box-shadow: 0 4px 15px rgba(0,0,0,0.1);
31     width: 90%;
32     max-width: 600px;
33     text-align: center;
34 }
35
36 /*timer*/
37 #timer-container {
38     font-size: 1.2rem;
39     font-weight: bold;
40     color: #d9534f;
41     margin-bottom: 1.5rem;
42     text-align: right;
43 }
```

```
44
45 /*question and answer*/
46 √ #question {
47     margin-bottom: 1.5rem;
48     color: #2c3e50;
49 }
50
51 √ .btn-grid {
52     display: grid;
53     grid-template-columns: repeat(2, 1fr);
54     gap: 15px;
55 }
56
57 √ .btn {
58     padding: 12px;
59     border: 2px solid #3498db;
60     border-radius: 8px;
61     background-color: #fff;
62     color: #3498db;
63     font-size: 1rem;
64     font-weight: 600;
65     cursor: pointer;
66     transition: all 0.3s ease;
67 }
68
69 √ .btn:hover:not([disabled]) {
70     background-color: #3498db;
71     color: #fff;
72 }
73
74 /*answer feedback*/
75 √ .btn.correct {
76     background-color: #2ecc71;
77     color: white;
78     border-color: #2ecc71;
79 }
```

```
80
81 .btn.incorrect {
82     background-color: #e74c3c;
83     color: white;
84     border-color: #e74c3c;
85 }
86
87 #feedback-section {
88     min-height: 1.5rem;
89     margin: 1.5rem 0;
90     font-weight: bold;
91     font-size: 1.1rem;
92 }
93
94 /*controls*/
95 .controls {
96     display: flex;
97     justify-content: center;
98     margin-top: 1rem;
99 }
100
101 #next-btn {
102     background-color: #3498db;
103     color: white;
104     padding: 12px 24px;
105     font-size: 1.1rem;
106 }
107
108 #next-btn:hover {
109     background-color: #2980b9;
110 }
111
112 /*footer*/
113 footer {
114     margin-top: 2rem;
115     font-size: 1.2rem;
116     font-weight: bold;
117     color: #333;
118 }
```

```
119
120 .hide {
121     display: none;
122 }
123
124 @media (max-width: 500px) {
125     .btn-grid {
126         grid-template-columns: 1fr;
127     }
128 }
```

script.js

```
JS script.js > [?] questions
1 //questions array
2 const questions = [
3   {
4     question: "What is the capital of France?",
5     options: ["Paris", "London", "Berlin", "Madrid"],
6     answer: 0 //paris
7   },
8   {
9     question: "Which planet is known as the Red Planet?",
10    options: ["Venus", "Jupiter", "Saturn", "Mars"],
11    answer: 3 //mars
12  },
13  {
14    question: "What is the largest mammal on Earth?",
15    options: ["African Elephant", "Blue Whale", "Giraffe", "Hippopotamus"],
16    answer: 1 //blue whale
17  },
18  {
19    question: "In which year did World War II end?",
20    options: ["1943", "1947", "1945", "1950"],
21    answer: 2 //1945
22  },
23  {
24    question: "Who wrote 'Romeo and Juliet'?",
25    options: ["Charles Dickens", "William Shakespeare", "Jane Austen", "Mark Twain"],
26    answer: 1 //william shakespeare
27  }
28 ];
29
30 const questionElement = document.getElementById('question');
31 const answerButtonsElement = document.getElementById('answer-buttons');
32 const nextButton = document.getElementById('next-btn');
33 const feedbackText = document.getElementById('feedback-text');
34 const timeElement = document.getElementById('time');
35 const scoreContainer = document.getElementById('score-container');
36 const scoreElement = document.getElementById('score');
37 const totalQuestionsElement = document.getElementById('total-questions');
38 const quizSection = document.getElementById('quiz-section');
39
```

```

40 let shuffledQuestions, currentQuestionIndex;
41 let score = 0;
42 let timerInterval;
43 let timeLeft;
44 const TIME_PER_QUESTION = 15; //time in seconds
45
46 nextButton.addEventListener('click', () => {
47   currentQuestionIndex++;
48   nextQuestion();
49 });
50
51 //start the quiz
52 startQuiz();
53
54 //startQuiz()
55 function startQuiz() {
56   score = 0;
57   currentQuestionIndex = 0;
58   scoreContainer.classList.add('hide');
59   quizSection.classList.remove('hide');
60   shuffledQuestions = shuffleQuestions([...questions]);
61   totalQuestionsElement.innerText = questions.length;
62   nextQuestion();
63 }
64
65 //shuffleQuestions()
66 function shuffleQuestions(array) {
67   for (let i = array.length - 1; i > 0; i--) {
68     const j = Math.floor(Math.random() * (i + 1));
69     [array[i], array[j]] = [array[j], array[i]];
70   }
71   return array;
72 }
73

```

```

74 //nextQuestion()
75 function nextQuestion() {
76   resetState();
77   if (currentQuestionIndex < shuffledQuestions.length) {
78     displayQuestion(shuffledQuestions[currentQuestionIndex]);
79     startTimer();
80   } else {
81     showScore();
82   }
83 }
84
85 //displayQuestion()
86 function displayQuestion(question) {
87   questionElement.innerText = question.question;
88   question.options.forEach((option, index) => {
89     const button = document.createElement('button');
90     button.innerText = option;
91     button.classList.add('btn');
92     button.dataset.index = index;
93     button.addEventListener('click', selectAnswer);
94     answerButtonsElement.appendChild(button);
95   });
96 }
97
98 //startTimer()
99 function startTimer() {
100   timeLeft = TIME_PER_QUESTION;
101   timeElement.innerText = timeLeft;
102   timerInterval = setInterval(() => {
103     timeLeft--;
104     timeElement.innerText = timeLeft;
105     if (timeLeft <= 0) {
106       clearInterval(timerInterval);
107       handleTimeOut();
108     }
109   }, 1000);
110 }

```

```

111
112 function handleTimeout() {
113     feedbackText.innerText = "Time's Up!";
114     feedbackText.style.color = "■ #e74c3c";
115     Array.from(answerButtonsElement.children).forEach(button => {
116         button.disabled = true;
117
118         if (parseInt(button.dataset.index) === shuffledQuestions[currentQuestionIndex].answer) {
119             button.classList.add('correct');
120         }
121     });
122     nextButton.classList.remove('hide');
123 }
124
125 //checkAnswer()
126 function selectAnswer(e) {
127     clearInterval(timerInterval);
128     const selectedButton = e.target;
129     const selectedIndex = parseInt(selectedButton.dataset.index);
130     const correctIndex = shuffledQuestions[currentQuestionIndex].answer;
131
132     if (selectedIndex === correctIndex) {
133         score++;
134         feedbackText.innerText = "Correct!";
135         feedbackText.style.color = "■ #2ecc71";
136         selectedButton.classList.add('correct');
137     } else {
138         feedbackText.innerText = "Incorrect!";
139         feedbackText.style.color = "■ #e74c3c";
140         selectedButton.classList.add('incorrect');
141
142         Array.from(answerButtonsElement.children).forEach(button => {
143             if (parseInt(button.dataset.index) === correctIndex) {
144                 button.classList.add('correct');
145             }
146         });
147     }

```

```

148
149     //disable all buttons after answer is selected
150     Array.from(answerButtonsElement.children).forEach(button => {
151         button.disabled = true;
152     });
153
154     nextButton.classList.remove('hide');
155 }
156
157 function resetState() {
158     clearInterval(timerInterval);
159     nextButton.classList.add('hide');
160     feedbackText.innerText = '';
161     while (answerButtonsElement.firstChild) {
162         answerButtonsElement.removeChild(answerButtonsElement.firstChild);
163     }
164 }
165
166 //display the final score
167 function showScore() {
168     quizSection.classList.add('hide');
169     scoreElement.innerText = score;
170     scoreContainer.classList.remove('hide');
171 }

```